

ANTARCTIC PENINSULA

The coldest, most remote continent is ruled by ice

More than 99 percent of Antarctica—Earth's driest, coldest, windiest continent—is covered by ice. Only the 1,240-mile (2,000-km) long Antarctic Peninsula reaches beyond the Antarctic Circle, pointing north toward Cape Horn. Life here is shaped by extremes. Antarctica's interior is elevated, with air so cold it cannot hold moisture, yet the coastal belt is damp. Cold air falls from the interior in blistering gales. Rain, fog, and blizzards alternate with sunny days when temperatures reach 41°F (5°C). Months of darkness give way to summers of 24-hour daylight, but even the best summer conditions are still challenging.

Life among the ice

The waxing and waning of sea ice is the driving force for most Antarctic life. Pack ice expands by up to 3 miles (4.8 km) per day, thickening to 7 ft (2.1 m), with fallen snow on top. For much of the year, ice locks away feeding

areas and breeding sites for many creatures. Most seals and penguins, skuas, and other birds only breed when the ice melts to reveal solid rock, so they move away from land as the ice expands. Emperor penguins, however, head south, where the males endure the worst winter conditions as they incubate eggs and fast for 65 days while the females go back to the ocean to feed. Weddell seals remain, using breathing holes so they can live under the ice all winter.

While approximately 300 species of algae, 200 lichens, 85 mosses, and 25 liverworts are known to exist in this icy landscape, only two flowering plant species are considered to be native to the Antarctic.



WHITE-BLOODED FISH
Blackfin icefish survive in temperatures that would freeze most other fish solid. Glycoproteins prevent their body fluids from freezing, and although their blood lacks hemoglobin (red blood cells), it is more fluid and so uses less energy to circulate.

BLACKFIN ICEFISH

SNOW PETREL

LOYAL MATE
Snow petrels nest farther south than any other bird, with many colonies along the peninsula. Mated pairs remain faithful for life, and lay one egg per year in a rock crevice. Their pure white plumage camouflages them against snow and ice.

ANTARCTIC PEARLWORT

EXPANDING BLOOMS
One of only two flowering plants on the entire continent, Antarctic pearlwort forms green cushions, seen more and more frequently on the Antarctic fringe. This white-blossomed plant is widening its range due to a warming climate.

So dry it is classified as a cold desert